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Effect of Time of Day on Animal Occupancy Near Water

Introduction

Wildlife management and viewing has been largely affected by the installment of artificial water points, especially in the arid conditions of the African savanna (Parker & Witkowski, 1999). From the 1960s to the 1990s, drought has led to the establishment of many artificial water points in the Kruger National Park and across Greater Kruger (Smit, 2013). However, since the 1990s, the consequences of their installment, such as overgrazing, starvation, rising predation, and herbaceous species composition, has led to the closure of many of these waterpoints in the Kruger National Park (Walker et al. 1987; Parker and Witkowski 1999; Smit 2013; Owen-Smith and Mills 2006). Derek Fedak, a PhD candidate at Colorado State University, is looking into the effects this management decision has had on animal habitat and corridor use through his research on seasonal animal occupancy, using camera traps placed along game trails. Transects of three camera traps were placed in different areas, including one near the Renosterkoppies Dam, in the Kruger National Park, during both dry and rainy seasons. The data discussed in this report is part of the data collected for Derek's research and were taken at this site, by camera B211105716, which is 25-50 m away from the waterpoint, at plot RKP2. Data was stored on SD card RK2, in folders 112EK113, 113EK113, and 114EK113. The images analyzed were taken from March 8 to 12 in 2023, over a period of five days.

Methods

At each site, sets of three camera traps were placed along a piospheric gradient: camera 1 was placed at the chosen waterpoint, camera 2 was placed 25-50 m away from the waterpoint, and camera 3 was placed 75-150 m away from the waterpoint, facing a game trail (Fig. 1). This transect was configured in this way to capture animal movement and changes in occupancy along the piospheric gradient. Camera traps were installed and maintained according to standard procedures for camera traps and the Camera Trap Field Protocol (Fig. 2; Fedak, 2023a). Each camera was set to capture an image at 1500 hours everyday as a control photo for vegetation and climate. In addition, cameras were set to capture a burst of three images, with one-second intervals between each image and a 30-second quiet period between bursts, when triggered by motion.

Figure 1. Approximate locations of camera trap plots placed at Renosterkoppies, as of May 12, 2023.

Figure 2. SD cards being collected and camera trap maintenance at plot RKP2 (left and center) and RKP3 (right).

For photo analysis, the number of individuals in each burst of three images was tallied, according to the Camera Trap Analysis Protocol (Fedak, 2023b). Exposure, brightness, contrast, and shadows on dark or blurry images were edited to enhance photo quality and improve accuracy when deemed helpful or necessary (Fig. 3). Species, time, date, site, camera number, and visit number, as well as sex (male, female) and age (adult, juvenile) if easily identifiable, were recorded. For data analysis, the number of animals recorded between each hour, e.g., between 800 and 900 hours, were summed and then compared.

Figure 3. Example of two images in the same burst, however the brightness and exposure was increased and the contrast was lowered in the image on the bottom.

Results

On most days, the maximum number of animals recorded fell between 1300 and 1500 hours, i.e., between 1 and 3 pm (Fig. 4). From 1300 to 1400 hours, there were the most cumulative number of animals recorded by the camera trap, across the five-day period. More specifically, on March 8, the highest number of animals, 183, were captured between 1300 and 1400 hours. On March 9, the highest was 313 animals, which appeared between 1400 and 1500 hours. On March 10, the highest amount was 66, between 1500 and 1600 hours. On March 11, a maximum of 944 animals were recorded between 1300 and 1400 hours. March 12, was somewhat of an outlier with the highest number of animals, 148, captured from 1700 to 1800 hours. There were also no herds recorded until 1500 hours, on March 12th and 10th, unlike other days when most herds arrived from 1000 to 1200 hours.

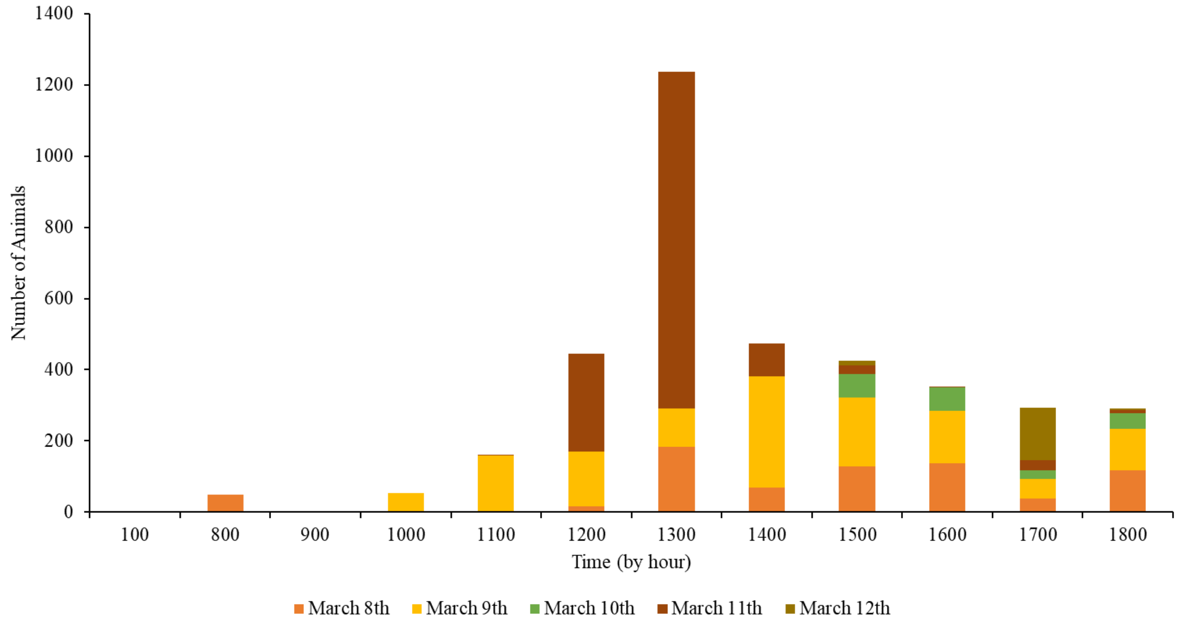

Figure 4. Number of animals at different times of day, where each column is the sum of animals recorded within the hour on different dates across a five-day period, and each hour represents animals seen between that time and the subsequent hour, e.g., 800 represents animals seen from 800 to 900 hours.

As for species composition, most species seen at plot RKP2 were grazers and browsers, including impala (*Aephyceros melampus*), greater kudu (*Tragelaphus strepsiceros*), warthogs (*Phacochoerus africanus*), zebra (*Equus quagga*), wildebeest (*Connochaetes taurinus*), and the African elephant (*Loxodonta africana*). The number of most of these species was highest between 1200 to 1400 hours, during the five days, in total. However, African elephants were generally seen passing by in darker conditions, such as during the nighttime or later in the day when it was rainy and overcast. Some bird species were recorded, including the red-billed oxpecker (*Buphagus erythrorynchus*) and blacksmith plover or blacksmith lapwing (*Vanellus armatus*). There were few blacksmith plovers but the trend in the number of red-billed oxpeckers seemed to follow the general trend in impalas but on a smaller scale.

Hour	impala	red-billed oxpecker	African elephant	blacksmith plover	zebra	kudu	warthog	wildebeest	Total
100			1						1
800	48								48
1000	48	5							53
1100	152	9							161
1200	382	17				40	6	1	446
1300	1114	26				96			1236
1400	469	2			3				474
1500	421	4		1					426
1600	351			1					352
1700	288		3	2					293
1800	392								392

Figure 5. Number of individuals in each species, seen between different hours over a period of five days, and the total number of animals recorded between each hour.

Discussion

The results support the hypothesis that more animals are present 25-50 m in proximity to the waterhole around midday. The highest number of animals were recorded between 1300 and 1500 hours on most days, over a period of five days. This could be because they group together in a less vulnerable area during nightfall, in order to avoid predation as most of the animals recorded are herbivores, with many large predators which are active at night (Hayward & Hayward, 2012). The number of animals did not seem directly correlated with changes in temperature. For example, on March 8, temperatures rose from 28 degrees C at approximately 1200 hours, up to 36 degrees C until approximately 1800 hours, while the number of animals was the highest between 1300 and 1400 hours and remained lower afterwards. However, their movement throughout the day may be partly due to the rising atmospheric temperature as the day progresses, which leads animals to move towards water points or rest in shadier, cooler areas, such as under the large tree in Fig. 3, and thus arrive near the waterpoint around this time, as weather and heat stress can affect the movement and activity of impalas and large antelopes (Kurauwone et al. 2013; Shrestha et al. 2014). Although, the question of where these herds go once they leave the RKP2's camera's view remains. Do they eventually arrive and stay at the waterpoint itself, or stay in the vicinity but move throughout the day to avoid predation? Likewise, the occasional and brief appearance of African elephants may indicate that they are nearby, but perhaps spend more time at the waterpoint (Purdon & Van Aarde, 2017). Analysis of images taken on the same day, from the camera traps at RKP1 and RKP3, would provide more information. In addition, red-billed oxpeckers were often perched on impalas, eating ticks and other insects, which may be the reasons that the trend in their abundance followed the general trend of impala abundance and other grazers and browsers, such as the greater kudu.

However, one note is that there may be some possible experimental errors including miscounting animals or misidentifying species during the image analysis, especially in low-light or faraway conditions, when there was a large number of animals in one image such as greater than ten, or when an animal was only partially visible in an image. Another potential issue was double-counting animals. Often, the same individual or herd would trigger the camera multiple times within several minutes, resulting in many bursts of the same individual or group of animals. This could make it seem as if there are more animals at a given time then there actually is, if one particular individual or herd remained in front of the camera, especially when camera traps are often placed by trees and bushes which provide food and shade. Thus, the supposed number of animals recorded is actually a representation of both the number of animals and the length of time they remained in the plot. It also appears that many of the photos taken over this five day period may have been triggered by moving grass or a branch. In some bursts of images, not including the daily photo, there were no animals walking directly in front of the camera and sometimes there appeared to be multiple bursts of photos with no wildlife present. However, due to the location of the camera with a wide view of a game trail and grassy field, although the camera may have been falsely triggered, there were often animals in the background.

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